# T3 Weinberg-operator coefficient comparison

## Notation key

| symbol | interaction / definition |
|---|---|
| $g_1$ | $D_\mu \supset i g_1 Y B_\mu$ |
| $g_2$ | $D_\mu \supset i g_2 T^A W_\mu^A$ |
| $g_3$ | $D_\mu \supset i g_3 t^A G_\mu^A$ |
| $Y_u$ | $-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$ |
| $Y_d$ | $-\bar{Q} Y_d H d + \text{h.c.}$ |
| $Y_e$ | $-\bar{L} Y_e H e + \text{h.c.}$ |
| $y_1$ | $L F S_1 + \text{h.c.}$ |
| $y_2$ | $L F S_2 + \text{h.c.}$ |
| $M_F$ | $-\bar{F} M_F F$ |
| $T$ | $\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$ | $\text{Tr}(y_1^\dagger y_1)$ |
| $T_\nu^{(2)}$ | $\text{Tr}(y_2^\dagger y_2)$ |
| $T_\nu$ | $T_\nu^{(1)} + T_\nu^{(2)}$ |
| $m_1^2$ | $m_1^2 S_1^\dagger S_1$ |
| $m_2^2$ | $m_2^2 S_2^\dagger S_2$ |
| $\lambda_1$ | $\frac{1}{2}\lambda_1 (H^\dagger H)^2$ |
| $\lambda_{S_1}^{(1)}$ | $\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$ |
| $\lambda_{S_2}^{(1)}$ | $\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$ |
| $\lambda_{HS_1}^{(1)}$ | $\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$ |
| $\lambda_{HS_2}^{(1)}$ | $\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$ |
| $\lambda_{HS_1}^{(A)}$ | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$ |
| $\lambda_{HS_2}^{(A)}$ | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$ |
| $\lambda_5$ | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$ |
| $\lambda_{12}^{(1)}$ | $\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$ |
| $\lambda_{12}^{(A)}$ | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$ |
| $\lambda_{12}^{(\times)}$ | $\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$ |
| $\lambda_{S_1}^{(A)}$ | $\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$ |
| $\lambda_{S_2}^{(A)}$ | $\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$ |
| $\alpha$ | $Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$ |
| $C_2(d)$ | $(d^2-1)/4$ |

| | |
|---|---|
| $T(d)$ | $d(d^2-1)/12$ |
| $C^{ij;AB}_{LLS_1S_2}$ | $L_iL_jS_1^AS_2^B$ |
| $C_{ijab}$ | $L_iL_j\phi_a\phi_b$ |
| $C_5^{ij}$ | $(L_i^TCL_j)HH$ |
| $R_W$ | $16\pi^2\beta_{C_5}\supset R_W\lambda_5 C_{LLS_1S_2}$ |

| Model | Ordered coefficient extracted from Matchete | Symmetric physical Wilson coefficient |
|---|---|---|
| T3-A-m4 | $A_5^{ij}=$ | $C_5^{ij}=$ |
| | $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ | $\frac{2}{3}+\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}\sqrt{3}\hbar y_{2,j,r}}^2\lambda_5^2m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-A-m2 | $A_5^{ij}=$ | $C_5^{ij}=$ |
| | $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ | $\frac{2}{3}+\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}\sqrt{3}\hbar y_{2,j,r}}^2\lambda_5^2m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-A-p0 | $A_5^{ij}=$ | $C_5^{ij}=$ |
| | $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ | $\frac{2}{3}+\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}\sqrt{3}\hbar y_{2,j,r}}^2\lambda_5^2m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-B-m3 | $A_5^{ij}=$ | $C_5^{ij}=$ |
| | $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ | $\frac{2}{3}+\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}\sqrt{3}\hbar y_{2,j,r}}^2\lambda_5^2m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-B-m1 | $A_5^{ij}=$ | $C_5^{ij}=$ |
| | $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ | $\frac{2}{3}+\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}\sqrt{3}\hbar y_{2,j,r}}^2\lambda_5^2m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-B-p1 | $A_5^{ij}=$ | $C_5^{ij}=$ |
| | $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ | $\frac{2}{3}+\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}\sqrt{3}\hbar y_{2,j,r}}^2\lambda_5^2m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-C-m3 | $A_5^{ij}=$ | $C_5^{ij}=$ |
| | $-\frac{2}{3}\left(\frac{\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ | $-\frac{2}{3}+\left(\frac{\hbar\overline{y_{1,i,r}\sqrt{3}\hbar y_{2,j,r}}^2\lambda_5^2m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-C-m1 | $A_5^{ij}=$ | $C_5^{ij}=$ |
| | $-\frac{2}{3}\left(\frac{\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ | $-\frac{2}{3}+\left(\frac{\hbar\overline{y_{1,i,r}\sqrt{3}\hbar y_{2,j,r}}^2\lambda_5^2m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |

| Model | Ordered coefficient extracted from Matchete | Symmetric physical Wilson coefficient |
|---|---|---|
| T3-C-p1 | $A_5^{ij} =$ | $C_5^{ij} =$ |
| | $-\frac{2}{3}\left(\frac{\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$ | $\frac{2}{3}\left(\frac{\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-D-m2 | $A_5^{ij} =$ | $C_5^{ij} =$ |
| | $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$ | $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-D-p0 | $A_5^{ij} =$ | $C_5^{ij} =$ |
| | $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$ | $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-D-p2 | $A_5^{ij} =$ | $C_5^{ij} =$ |
| | $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$ | $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-E-m4 | $A_5^{ij} =$ | $C_5^{ij} =$ |
| | $-\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$ | $\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-E-m2 | $A_5^{ij} =$ | $C_5^{ij} =$ |
| | $-\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$ | $\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-E-p0 | $A_5^{ij} =$ | $C_5^{ij} =$ |
| | $-\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$ | $\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |
| T3-E-p2 | $A_5^{ij} =$ | $C_5^{ij} =$ |
| | $-\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$ | $\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$ |